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RESEARCH

Clinical outcomes of low versus high intensity laser therapy in patients with subacute carpal tunnel syndrome: A randomized, single-blinded, controlled trial

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Abstract

A common entrapment of median nerve that causes pain, numbness, and functional impairment is called Carpal Tunnel Syndrome (CTS). Laser therapy is a noninvasive intervention for neural repair and pain modulation, but comparative evidence between Low-Level Laser Therapy (LLLT) and High-Intensity Laser Therapy (HILT) in subacute CTS is limited. The aims of this study were to compare the effects of LLLT and HILT on pain, functional status, electrophysiological, and ultrasonographic parameters in subacute CTS. A single-blinded randomized controlled trial (Reg. No. IRCT20241111063665N1) was conducted at the Department of Physiotherapy, The University of Lahore (UOL), Lahore, Pakistan, from August 2024 to March 2025. Sixty-nine participants with subacute CTS, confirmed by electromyography and nerve conduction studies, were randomly assigned to three groups ($n=23$ each): Group A (HILT; 808 nm + physiotherapy), Group B (LLLT; 860 nm + physiotherapy), and Group C (physiotherapy + sham laser). Assessments were at baseline and week 6. Outcomes included pain (VAS), symptom severity and functional status (Boston Carpal Tunnel Questionnaire; BCTQ), grip strength, distal motor latency (DML), sensory nerve conduction velocity (SNCV), and median nerve cross-sectional area (CSA). ANCOVA with Bonferroni correction and effect size estimation was applied. At week 6, both HILT and LLLT showed significant improvements ($p<0.001$). HILT achieved greater pain reduction (−4.9) than LLLT (−4.1) and control (−2.2; $p<0.001$), with greater improvement in BCTQ-SSS (−2.2 points), BCTQ-FSS (−1.9 points), grip strength (+9.9 kg), DML (−1.2 ms), SNCV (+12.3 m/s), and CSA (−3.7 mm²). Large effect sizes (Cohen's $d>0.8$) were observed. Both LLLT and HILT are effective as adjuncts to physiotherapy in subacute CTS. HILT demonstrated greater improvements than LLLT within the parameters tested.
Clinical Trial Number: IRCT20241111063665N1, Registration date: 17 March, 2025.

Keywords Carpal Tunnel Syndrome · Laser Therapy · Low-Level Laser Therapy · Pain · Nerve Conduction Studies · Ultrasonography

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Introduction

Carpal tunnel syndrome (CTS) is the most common type of peripheral nerve entrapment caused by compressing the median nerve within the wrist. Symptoms of CTS include pain, numbness, and tingling, and may cause a loss of function and grip strength in the hand, particularly at night, and negatively impact one's level of daily activities and quality of life [1]. Estimates of CTS prevalence vary considerably around the world; a meta-analysis of over 5.3 million subjects reported an overall estimate of approximately 14.4% with higher prevalence rates in high-income countries (16.9%) than in low- and middle-income countries (11.4%). Previous population-based studies have reported a prevalence of 14% to 16% [2]. Certain occupational exposures such as repeated use of the hands and wrists, awkward postures, or sustained grip have also been consistently associated with increased risk of developing CTS across all the different populations and work environments [3].

Conservative management of CTS remains the first-line treatment for mild to moderate cases of CTS [2]. Treatment typically involves the use of splints, modification of activities, exercises, and various forms of physical modalities [4]. Among such modalities, laser therapy has received substantial attention due to its anti-inflammatory and analgesic potential as well as its noninvasive nature [5]. Low-level laser therapy (LLLT) is one form of photobiomodulation, whereas, high-intensity laser therapy (HILT) has a multitude of different protocols for clinical use; some differences include variations in wavelength, power output and energy delivery which may warrant different biological effects, including the modulation of pain or the penetration of tissue [6].

Both HILT and LLLT are forms of photobiomodulation that are believed to promote tissue repair and modulate pain through cellular and biochemical mechanisms. Laser energy is absorbed by mitochondrial chromophores, resulting in increased adenosine triphosphate production, modulation of inflammatory mediators, and enhancement of microcirculation. In CTS, these effects may contribute to reduced median nerve inflammation and edema, improved nerve conduction, and alleviation of pain and functional impairment. HILT may additionally achieve greater tissue penetration and energy delivery, potentially producing more pronounced biological effects than LLLT [6].

The scientific evidence supporting the use of laser therapies in the treatment of CTS is expanding yet remains inconsistent [7]. Some randomized controlled trials and systematic reviews suggest that HILT produces greater decreases in pain and disability, as well as improved electrophysiological measurements when used alone or in conjunction with other treatments such as splinting, exercise, and ergonomic modifications [8–10]. In contrast, the literature supporting LLLT is mixed with typically poor-quality data (and in many cases applying large sample sizes), as well as variable, inconclusive results for symptom severity, functional status and/or electrophysiology, short follow-up periods and/or methodological heterogeneity [11]. Very few studies have directly compared HILT to LLLT, and only a small number of studies have specifically recruited patients with subacute CTS prior to developing chronic nerve dysfunction [12].

There is a paucity of evidence that assists clinicians with choosing the best laser treatment modality and protocol/dosage of therapy for patients with subacute CTS, in which timely management might have prevented patients from progressing to chronic disability. Additionally, there exists a significant lack of standardized patient reported outcome measures being utilized in most studies conducted to date [13–15]. Although previous studies have reported beneficial effects of both HILT and LLLT in patients with CTS, direct comparisons between these modalities remain limited. Furthermore, most available studies have focused on general CTS populations rather than individuals with subacute CTS, a stage in which early intervention may influence disease progression and recovery. Consequently, this study aims to compare the clinical, electrophysiological, and ultrasonographic outcomes between patients treated with LLLT and HILT for subacute CTS to fill the knowledge gaps and provide relevant evidence to practice physiotherapy.

Patients and methods

This randomized, single-blinded, parallel-group clinical trial was conducted at the Department of Physical Therapy, The University of Lahore (UOL), Lahore, Pakistan. Institutional Ethics Review Board of the University (Approval No. REC-UOL-576/08/24), from August 2024 to March 2025, adhering to the CONSORT 2010 statement and its extension for non-pharmacologic interventions and the TIDieR checklist for intervention reporting. The study protocol has been registered retrospectively with the Iranian Registry of Clinical Trials (Reg. No.: IRCT20241111063665N1, Registration date: 17 March, 2025). Participant recruitment commenced on 29 August 2024, and continued through March 2025.

G*Power software (version 3.1.9.2) was used for a sample size of 69 participants ($n = 23$, per group) based on the primary outcome (Boston Carpal Tunnel Questionnaire; BCTQ), with a variance of 0.64, a significance level (α) of 0.05, and 80% power [16]. No adjustment for anticipated attrition was applied in the sample size calculation. A non-probability convenience sampling technique was used to recruit participants meeting eligibility criteria and providing informed consent.

Inclusion criteria included age 20–40 years, mild subacute CTS confirmed by electromyography and nerve conduction studies (sensory delay > 3.6 ms, motor delay > 4.1 ms), and involvement of both sensory and motor fibers. The subacute CTS was symptomatic for approximately 6–16 weeks (as this represents the early symptomatic phase of CTS, prior to development into chronic degenerative disease). The age range was selected to reduce heterogeneity associated with age-related degenerative changes and variations in peripheral nerve function that could influence treatment outcomes, thereby improving internal validity in a subacute CTS cohort. Exclusion criteria were no physical therapy or pain/functional outcome medications were utilized for 4 weeks and 2 weeks, respectively (no previous therapies to prevent complications due to prior treatments). Individuals with systemic or neurological conditions known to directly affect peripheral nerve function or conduction (e.g., hypothyroidism, myopathy, and myelopathy) were excluded to minimize bias in electrophysiological and ultrasonographic measures. Exclusions for potential safety and rehabilitation response ramifications or impact of neurophysiology included: malignancy, active infection, pulmonary disease, immunodeficiency, history of shoulder or neck surgery, corticosteroid use, drug addiction and/or pregnancy at high risk. In contrast, common comorbid conditions such as diabetes mellitus and hypertension were not excluded, as they are frequently present in patients with CTS.

Participants were randomly allocated into three groups using block randomization (block sizes of 4 and 6). Allocation sequence was generated and managed by an independent researcher not involved in recruitment or assessment and was not accessible to the investigators. Allocation was concealed using sequentially numbered, opaque sealed envelopes, and participants were assigned after baseline assessment. All groups received standardized physiotherapy exercises for 6 weeks (18 sessions total, 3 sessions/week). Group A received HILT combined with physiotherapy, Group B received LLLT combined with physiotherapy, and Group C served as a control group receiving physiotherapy combined with a sham laser application. The CONSORT flow diagram illustrating participant enrollment, randomization, follow-up, and analysis is presented in Fig. 1. To achieve high participant retention, close supervision of scheduled physiotherapy sessions, appointment-based treatment delivery, and a short trial duration were implemented.

The outcome assessors were blinded to group allocation. Due to the nature of the intervention, therapists delivering laser therapy were not blinded. Participants were blinded to group allocation through the use of an identical sham laser procedure. Accordingly, the study was conducted as a single-blind randomized controlled trial.

In this study, HILT was delivered using the Pagani Laser Scanner (Italy) at power output of 4.6 W, a wavelength of 808 nm, and 10 Hz frequency, and a manufacturer-specified spot size of 0.8 cm^2, yielding a calculated nominal irradiance of 5.75 W/cm^2. The laser was applied in contact mode using a stationary technique over ten standardized points along the median nerve and flexor retinaculum. The anatomical locations of the ten treatment points are illustrated in Fig. 2. The treatment points were selected according to the anatomical course of the median nerve within the carpal tunnel region and were standardized across all participants based on previously published laser therapy protocols for CTS [16, 17]. Each point received a target fluence of 48 J/cm^2 per session (equivalent to 38.4 J per point, totaling 384 J delivered across all 10 points per session). The beam was applied perpendicular (90° angle) to the skin surface with light, stable contact pressure over a total treatment time of 10 min per session. The device was manufacturer calibrated prior to the test (no independent verification external to the manufacturer of the device was conducted prior to undertaking the study). The selected treatment parameters were based on published CTS laser therapy protocols and were consistent with World Association for Photobiomodulation Therapy (WALT) recommendations for photobiomodulation therapy [16].

LLLT was delivered using the Lasermed 4098 device (Brazil) with a wavelength of 860 nm, power output of 50 mW (0.05 W), frequency of 10 Hz, and a manufacturer-specified spot size of 2.5 cm^2, yielding a calculated nominal irradiance of 0.02 W/ cm^2 (20 mW/cm^2). Application was performed at the same ten anatomical points in stationary mode with a target fluence of 12 J/cm^2 per point per session (equivalent to 30 J per point, totaling 300 J delivered across all 10 points per session). The probe was held perpendicular (at 90° angle) to the skin with minimal contact pressure over a 10-minute session duration. Calibration of delivery was to be completed according to the manufacturer's standards prior to the performance of the actual intervention. The selected parameters were based on published CTS laser therapy protocols and were consistent with WALT recommendations for photobiomodulation therapy [17].

Irradiance and beam profile were not directly measured during treatment sessions using an optical power meter, and reported spot sizes were based entirely on manufacturer specifications without independent external verification prior to or during the study. These values represent nominal parameters calculated from manufacturer-provided specifications, which should be taken into consideration regarding precise optical dose reproducibility. Both therapies are applied in a standardised, fixed point based, stationary technique, and not scanned technique, to ensure participants received the standardised amount of dosimetry for both therapies.

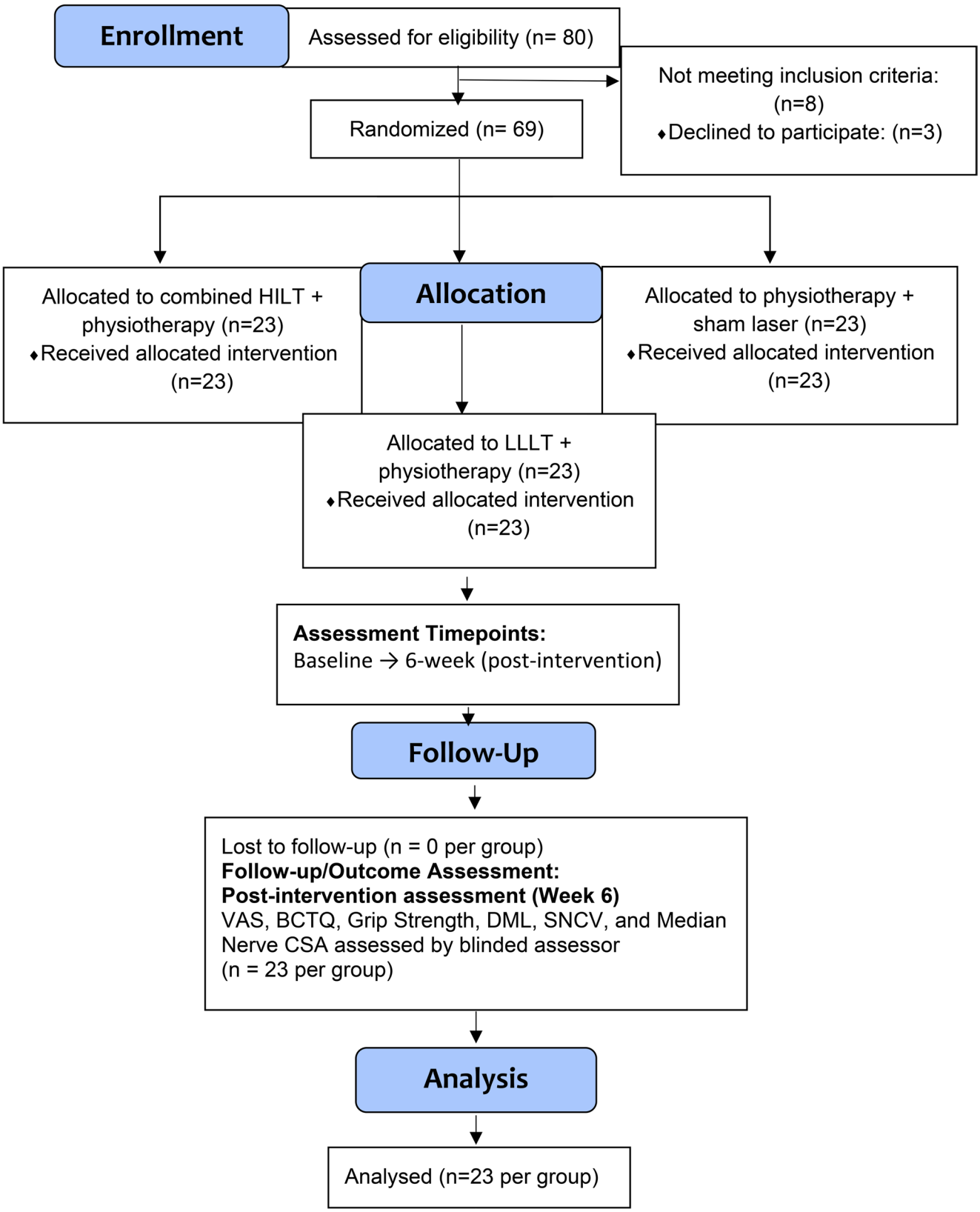


Fig. 1 CONSORT 2010 Flow Diagram

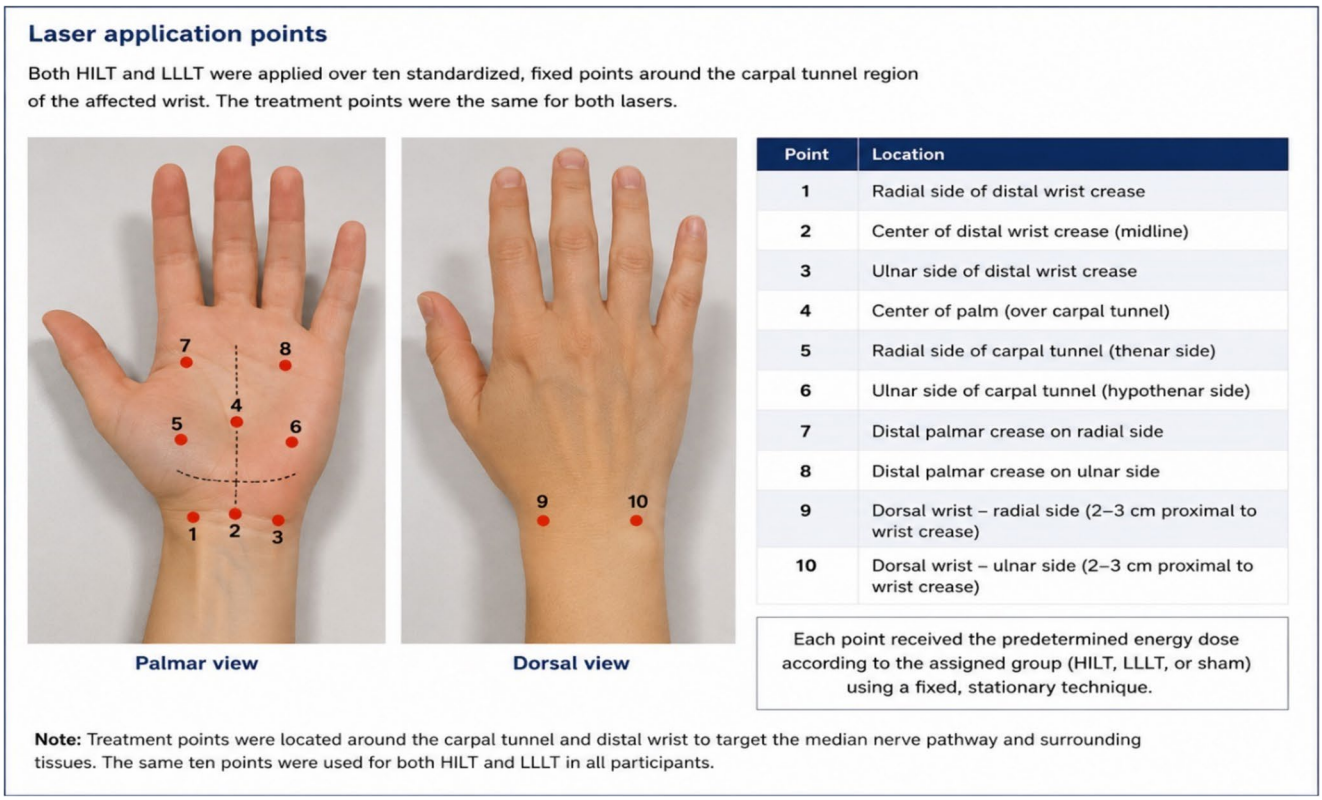


Fig. 2 Standardized anatomical treatment points used for HILT and LLLT in patients with subacute CTS

The sham intervention device served as an active laser treatment, providing equivalent visible light and sound as with the real device and was applied with identical positioning, duration, and procedural steps. The sham treatment received by the control group was provided an active sham laser device, with the same point of delivery; however, there was no therapeutic energy output from the sham laser.

All three treatment groups received a standardized rehabilitation program of active and active-assisted wrist extension, finger flexion and extension, as well as, weight bearing exercises. The physiotherapist that supervised the therapy provided a standardised exercise program to all three treatment groups. Each session consisted of approximately 20–25 min of the following exercises: (1) active wrist extension exercises (3 sets of 10 repetitions), (2) active-assisted wrist extension exercises (3 sets of 10 repetitions), (3) finger flexion and extension exercises (3 sets of 10 repetitions), and (4) weight-bearing exercises through the wrist (3 sets of 8–10 repetitions). A rest period of 30–60 s was taken between sets. Exercise intensity and range were progressed gradually based on patient tolerance and symptom response. All sessions were conducted three times per week over a six-week period. Adherence was monitored using physiotherapy logs, and participants who missed over 20% of sessions were flagged for sensitivity analysis. The physiotherapy session was administered after the laser intervention in the HILT and LLLT groups and after the sham therapy in the control group to maintain standardization of treatment sequencing across all groups.

The following outcomes were primary to the study: pain intensity (Visual Analogue Scale (VAS)) and functional status (BCTQ), which includes the two subscales of Symptom Severity and Functional Status). These were considered co-primary outcomes for the purpose of this study. Secondary outcomes included: grip strength (measured using a Jamar Hydraulic Dynamometer with participants seated, shoulder adducted, elbow flexed to 90°, forearm in neutral position, and wrist in slight extension according to standardized testing recommendations. Three consecutive measurements were obtained, and the mean value was used for analysis. The Jamar dynamometer has demonstrated excellent reliability and validity for hand strength assessment in patients with upper extremity disorders, including CTS [18]. Electrophysiological parameters (distal motor latency and sensory nerve conduction velocity) were performed under standardized laboratory conditions by an experienced examiner. Distal motor latency (DML) and sensory nerve conduction velocity (SNCV) of the median nerve were recorded using standard electrophysiological techniques. These measures are widely accepted objective indicators of median nerve dysfunction and are commonly used in the diagnosis and monitoring of CTS [19]. Sonographic assessment of the

cross-sectional area (CSA) of the median nerve was performed using a high-frequency linear transducer by an experienced sonographer. The CSA of the median nerve was measured at the level of the carpal tunnel inlet using the direct tracing method. Sonographic measurement of median nerve CSA has demonstrated good reliability and diagnostic validity in patients with CTS [20]. All assessments of primary and secondary outcomes were done by a blinded assessor at both baseline and following intervention completion, under standardized conditions.

Adverse events were defined as any unwanted or harmful reaction during or after treatment sessions, including pain aggravation, skin irritation, burns, dizziness, or neurological worsening, and were actively monitored at each treatment session through direct participant questioning and clinical observation by the treating physiotherapist.

Data analyses were carried out using SPSS version 26. Normality was tested via the Shapiro-Wilk test. Continuous variables are reported as mean ± SD while categorical variables are reported as counts/frequencies (%). Baseline comparisons were made using one way ANOVA or chi-square tests. Intervention effects were evaluated via ANOVA for covariates (ANCOVA) with post-intervention outcome measures as dependent variables; treatment groups as fixed factors; and baseline outcome measures to serve as covariates. Assumptions of ANCOVA, including homogeneity of regression slopes, linearity between covariates and outcomes, and normality of residuals, were assessed prior to analysis and were satisfied. Each outcome variable was analysed separately using ANCOVA models, and no interaction terms were included in the final models. Pairwise comparisons were adjusted via a Bonferroni correction procedure. Effect sizes were reported as Cohen's d and partial η^2. Cohen's d was calculated using the difference in post-intervention means between groups divided by the pooled standard deviation. A level of statistical significance of $p < 0.05$ was used to determine if there was an effect of intervention. Complete-case analysis was performed as there were no missing outcome data and no loss to follow-up. Therefore, imputation techniques were not required.

Results

A total of 69 participants with subacute CTS were randomly assigned to three different treatment groups: Group A (HILT + physiotherapy), Group B (LLLT + physiotherapy), and Group C (physiotherapy + sham laser), with 23 participants in each group. All randomized participants completed the intervention and post-intervention assessments, resulting in a 100% retention rate with no loss to follow-up (Fig. 1). The average age of participants was 34.2 ± 6.9 years with 59.4% being female. Baseline demographic and clinical characteristics were comparable between groups (Table 1).

Pain intensity (VAS) decreased significantly in all groups after six weeks of intervention ($p < 0.001$). Group A (HILT + PT) achieved the greatest reduction among the tested intervention protocols, followed by Group B (LLLT + PT) and Group C (PT + Sham Laser), with ANCOVA confirming a significant between-group difference ($F(2, 65) = 28.47$, $p < 0.001$, Partial $\eta^2 = 0.467$) (Table 2).

Figure 3 illustrates the mean change in pain intensity (VAS) from baseline to post-intervention across the three groups, allowing easier comparison of group differences that are summarized numerically in Table 2.

Functional status and symptom severity (BCTQ) improved significantly in all groups ($p < 0.001$), with the largest gains observed in the HILT protocol group. The Symptom Severity Scale (SSS) and Functional Status Scale (FSS) showed statistically significant between-group differences (SSS: $F(2,65) = 22.68$, $p < 0.001$, partial $\eta^2 = 0.411$; FSS: $F(2,65) = 19.92$, $p < 0.001$, partial $\eta^2 = 0.379$), indicating that the group receiving HILT had greater clinical recovery (Table 3).

Table 1 Patient's demographic and clinical characteristics, ($n = 69$)

Variable	Group A (HILT + PT), $n = 23$	Group B (LLLT + PT), $n = 23$	Group C (PT + Sham Laser), $n = 23$	p-value
Age (years), mean ± SD	34.6 ± 6.5	33.9 ± 6.9	34.1 ± 7.2	0.92
Gender (Male/Female)	9 / 14	10 / 13	9 / 14	0.94
BMI (kg/m²), mean ± SD	26.9 ± 3.1	26.4 ± 3.6	27.0 ± 3.5	0.81
Duration of Symptoms (weeks), mean ± SD	11.1 ± 2.9	11.5 ± 3.2	11.3 ± 3.1	0.89
Side of CTS (Right/Left)	14 / 9	13 / 10	15 / 8	0.83
Diabetes, n (%)	5 (21.7%)	4 (17.4%)	5 (21.7%)	0.89
Hypertension, n (%)	6 (26.1%)	5 (21.7%)	7 (30.4%)	0.77
HDL Cholesterol (mg/dL), mean ± SD	44.8 ± 6.9	45.1 ± 7.4	43.9 ± 7.3	0.87
Fasting Blood Glucose (mg/dL), mean ± SD	107.2 ± 17.6	105.9 ± 18.9	107.4 ± 19.1	0.94
Occupational Exposure Level; Low (< 2 h/day) Moderate (2–6 h/day) High (> 6 h/day)	6 (26.1%) 10 (43.5%) 7 (30.4%)	5 (21.7%) 11 (47.8%) 7 (30.4%)	7 (30.4%) 9 (39.1%) 7 (30.4%)	0.73

Table 2 Comparison of pain intensity (VAS) scores within and between groups

Group	Baseline	Post-intervention	Mean change (Δ)	Within-group *p*-value	Between-group comparisons (Post-hoc)	Adjusted *p*-value	Effect Size (Cohen's d)
Group A (HILT + PT)	6.8 ± 1.0	1.9 ± 0.9	−4.9	< 0.001	A vs. B	0.002	0.95 (Large)
Group B (LLLT + PT)	6.9 ± 1.1	2.8 ± 1.0	−4.1	< 0.001	A vs. C	< 0.001	2.59 (Very large)
Group C (PT + Sham Laser)	6.7 ± 1.2	4.5 ± 1.1	−2.2	< 0.001	B vs. C	< 0.001	1.62 (Very large)
p-value (between-groups)	0.88	< 0.001	-	-	-	-	-

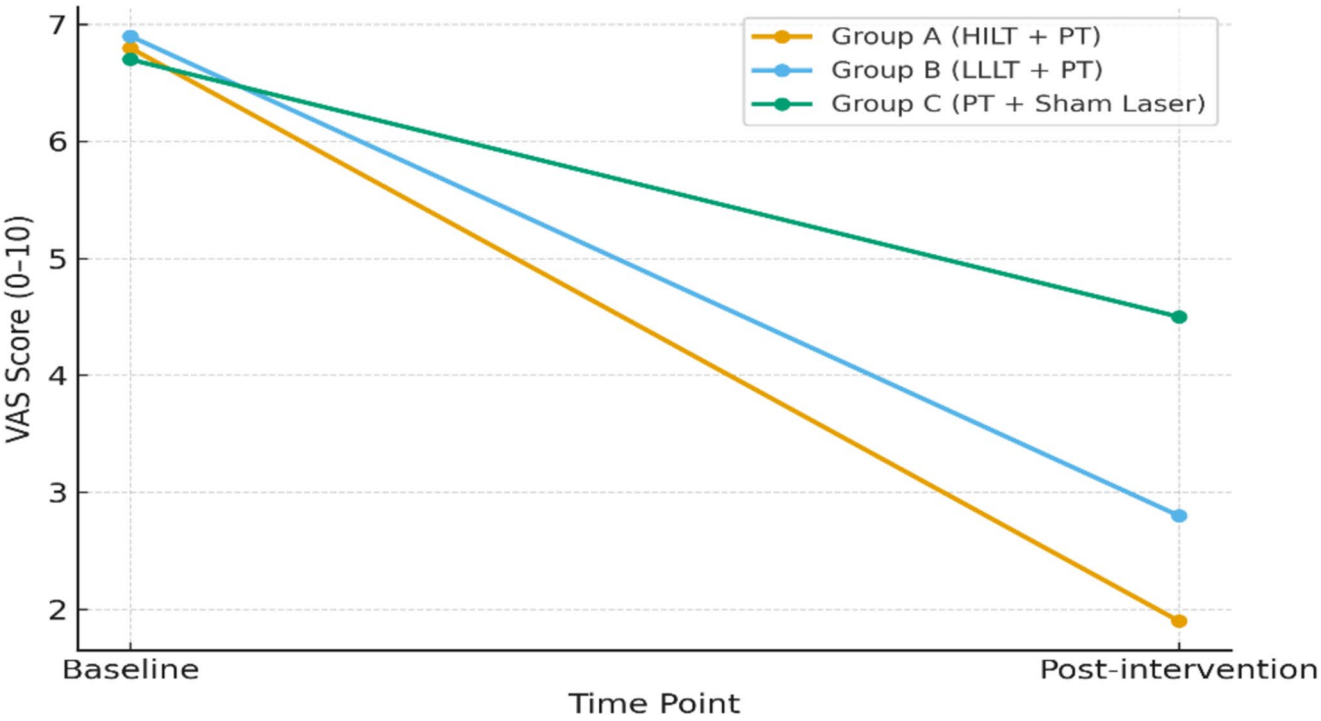


Fig. 3 Changes in pain intensity (VAS) across groups

Table 3 BCTQ symptom severity and functional status scores by groups comparison

Group	Baseline	Post-intervention	Mean change (Δ)	Within-group *p*-value	Between-group comparisons (Post-hoc)	Adjusted *p*-value	Effect Size (Cohen's d)
BCTQ – Symptom Severity Scale (SSS)							
Group A (HILT + PT)	3.9 ± 0.5	1.7 ± 0.4	−2.2	< 0.001	A vs. B	0.011	1.72 (Very large)
Group B (LLLT + PT)	3.8 ± 0.6	2.1 ± 0.5	−1.7	< 0.001	A vs. C	< 0.001	2.48 (Very large)
Group C (PT + Sham Laser)	3.7 ± 0.5	2.9 ± 0.6	−0.8	0.004	B vs. C	0.003	1.35 (Very large)
p-value (between-groups)	0.79	< 0.001	-	-	-	-	-
BCTQ – Functional Status Scale (FSS)							
Group A (HILT + PT)	3.4 ± 0.4	1.5 ± 0.3	−1.9	< 0.001	A vs. B	0.018	1.56 (Large)
Group B (LLLT + PT)	3.3 ± 0.5	1.9 ± 0.4	−1.4	< 0.001	A vs. C	< 0.001	2.25 (Very large)
Group C (PT + Sham Laser)	3.3 ± 0.4	2.6 ± 0.5	−0.7	0.006	B vs. C	0.009	1.12 (Large)
p-value (between-groups)	0.73	< 0.001	-	-	-	-	-

Grip strength increased significantly in all groups ($p < 0.01$). Between-group ANCOVA indicated a significant difference ($F(2,65) = 27.46$, $p < 0.001$; partial $\eta^2 = 0.458$), with post-hoc Bonferroni analysis showing greater improvement in Group A compared with Group B ($p = 0.032$) and Group C ($p < 0.001$) (Table 4).

Electrophysiological parameters improved in both HILT and LLLT groups, including reduced distal motor latency and increased sensory nerve conduction velocity. The HILT group exhibited the most pronounced changes, with significant between-group differences confirmed by ANCOVA for both DML ($F(2, 65) = 31.72$, $p < 0.001$,

Table 4 Grip strength (kg) for each group comparison

Group	Baseline	Post-intervention	Mean change (Δ)	Within-group *p*-value	Between-group comparisons (Post-hoc)	Adjusted *p*-value	Effect Size (Cohen's d)
Group A (HILT + PT)	17.9 ± 2.3	27.8 ± 2.6	+9.9	< 0.001	A vs. B	0.032	1.84 (Very large)
Group B (LLLT + PT)	18.1 ± 2.1	25.4 ± 2.9	+7.3	< 0.001	A vs. C	< 0.001	2.52 (Very large)
Group C (PT + Sham Laser)	18.0 ± 2.4	21.2 ± 2.8	+3.2	0.007	B vs. C	0.011	1.28 (Large)
p-value (between-groups)	0.85	< 0.001	-	-	-	-	-

Partial $\eta^2 = 0.494$) and SNCV ($F(2, 65) = 29.85$, $p < 0.001$, Partial $\eta^2 = 0.478$) (Table 5).

Median nerve CSA decreased significantly in all groups, with the greatest reduction in the HILT group and a significant between-group difference ($F(2, 65) = 28.96$, $p < 0.001$, Partial $\eta^2 = 0.471$) (Table 6).

Figure 4 shows the change in median nerve CSA across the three groups over the study period.

Adverse events were monitored during treatment sessions. No adverse events or complications were reported with either HILT or LLLT.

Discussion

This study compared clinical and neurophysiological outcomes of both HILT and LLLT in combination with routine physiotherapy for patients suffering from subacute CTS. The primary finding was that all three groups (the HILT, LLLT, and control group) experienced statistically significant improvement across the parameters of pain, function, grip strength, electrophysiology, and ultrasonography; however, the magnitude of improvement of the HILT group was greater than the LLLT group and the control group.

Statistically significant improvements in pain intensity were noted post-treatment in all three groups. The post-treatment VAS score decreased from 6.8 ± 1.0 to 1.9 ± 0.9 in the HILT group; 6.9 ± 1.1 to 2.8 ± 1.0 in the LLLT group; and 6.7 ± 1.2 to 4.5 ± 1.1 in the control group ($p < 0.001$). These findings are consistent with Alayat et al., who likewise found HILT produced greater reduction of pain compared with LLLT and placebo [21]. Reduced inflammatory mediators and increased production of endorphins may potentially explain the more effective pain modulation observed with HILT, although these mechanisms were not directly measured in the present study [22].

The BCTQ demonstrated significant results for patients treated with laser therapy in reference to SSS and FSS. This study found that the patients receiving HILT demonstrated the greatest improvements, with mean pre- and post-treatment SSS scores of 3.9 ± 0.5 and 1.7 ± 0.4, respectively, and FSS scores of 3.4 ± 0.4 and 1.5 ± 0.3, respectively. In the LLLT group, SSS scores changed from 3.8 ± 0.6 to 2.1 ± 0.5

Table 5 Electrophysiological outcome variables for each group comparison

Group	Baseline	Post-intervention	Mean change (Δ)	Within-group *p*-value	Between-group comparisons (Post-hoc)	Adjusted *p*-value	Effect Size (Cohen's d)
Distal Motor Latency (ms)							
Group A (HILT + PT)	5.3 ± 0.5	4.1 ± 0.4	−1.2	< 0.001	A vs. B	0.041	1.98 (Very large)
Group B (LLLT + PT)	5.3 ± 0.4	4.5 ± 0.4	−0.8	< 0.001	A vs. C	< 0.001	2.46 (Very large)
Group C (PT + Sham Laser)	5.4 ± 0.5	4.9 ± 0.5	−0.5	0.006	B vs. C	0.019	1.31 (Large)
p-value (between-groups)	0.69	< 0.001	-	-	-	-	-
Sensory Nerve Conduction Velocity (m/s)							
Group A (HILT + PT)	34.2 ± 3.6	46.5 ± 3.8	+12.3	< 0.001	A vs. B	0.028	2.02 (Very large)
Group B (LLLT + PT)	33.8 ± 3.9	43.9 ± 3.7	+10.1	< 0.001	A vs. C	< 0.001	2.55 (Very large)
Group C (PT + Sham Laser)	34.1 ± 3.5	38.3 ± 3.4	+4.2	0.004	B vs. C	0.017	1.36 (Large)
p-value (between-groups)	0.82	< 0.001	-	-	-	-	-

Table 6 Ultrasonographic outcomes (median nerve cross-sectional area, mm^2) for each group comparison

Group	Baseline	Post-intervention	Mean change (Δ)	Within-group *p*-value	Between-group comparisons (Post-hoc)	Adjusted *p*-value	Effect Size (Cohen's d)
Group A (HILT + PT)	14.6 ± 1.8	10.9 ± 1.4	−3.7	< 0.001	A vs. B	0.032	2.04 (Very large)
Group B (LLLT + PT)	14.4 ± 1.7	11.7 ± 1.5	−2.7	< 0.001	A vs. C	< 0.001	2.51 (Very large)
Group C (PT + Sham Laser)	14.5 ± 1.9	13.1 ± 1.7	−1.4	0.008	B vs. C	0.021	1.38 (Large)
p-value (between-groups)	0.93	< 0.001	-	-	-	-	-

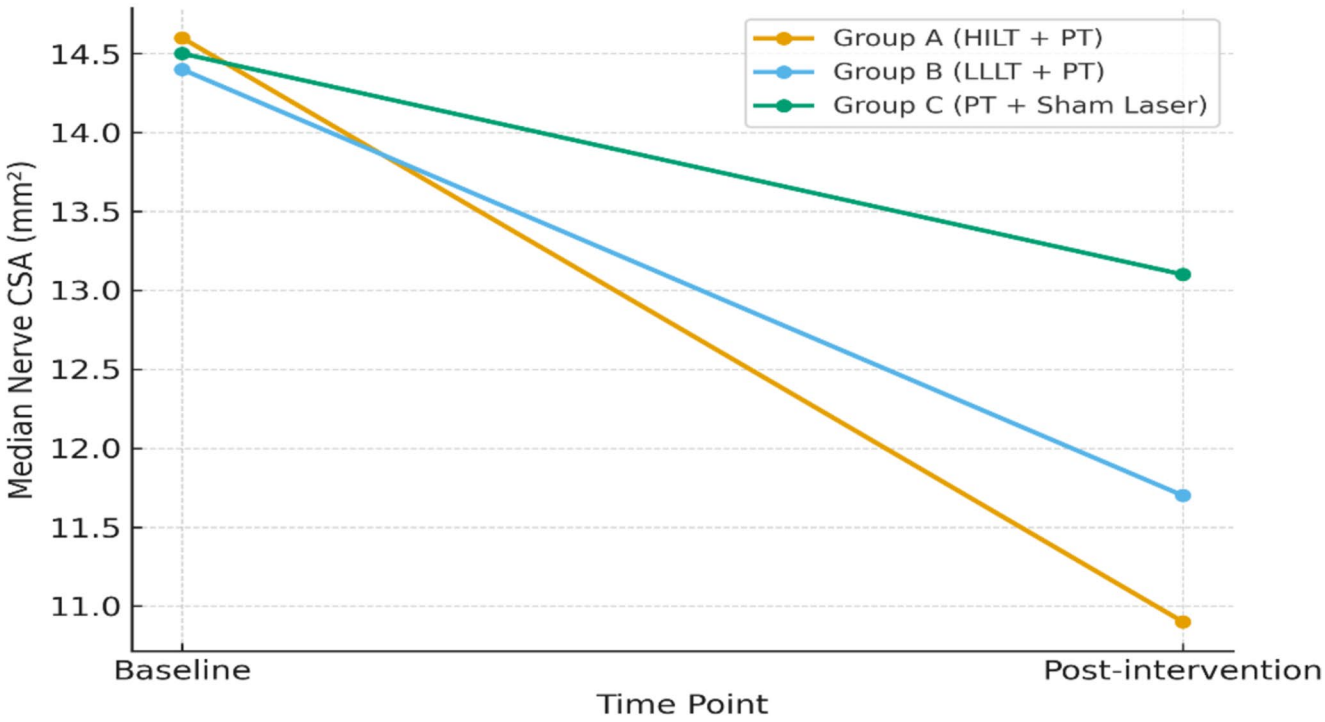


Fig. 4 Ultrasonographic changes in median nerve CSA

and FSS scores changed from 3.3 ± 0.5 to 1.9 ± 0.4. In the control group, SSS scores changed from 3.7 ± 0.5 to 2.9 ± 0.6 and FSS scores changed from 3.3 ± 0.4 to 2.6 ± 0.5. The patients receiving LLLT demonstrated moderate improvements, while the control group showed mild improvements. These findings are consistent with those of Yiğit and Ordahan who found that HILT provided superior functional recovery over LLLT [16]. The patients receiving HILT may have obtained better functional gains because of improved microvascular perfusion and the higher fluence delivered by the HILT protocol, as suggested in previous photobiomodulation studies, although these mechanisms were not directly assessed in the present study [21, 23].

In addition to statistical significance, the observed improvements appear to be clinically meaningful. Recent evidence suggests that the minimal clinically important difference (MCID) for pain intensity in patients with CTS is approximately 1.5 points, while the MCID values for the BCTQ-FSS and SSS are approximately 0.23 and 0.64 points, respectively [24]. In the present study, improvements in VAS, BCTQ-SSS, and BCTQ-FSS exceeded these thresholds in both laser groups, particularly in the HILT group, suggesting that the observed benefits were not only statistically significant but also clinically relevant to patients.

Additionally, both groups (HILT and LLLT) demonstrated significant increases in grip strength after treatment (9.9 kg increase in HILT group; 7.3 kg increase in LLLT group; 3.2 kg increase in control group). These findings are in line with Ashour et al. and Kim et al., who reported similar improvements following HILT [8, 25]. Improved grip strength may reflect enhanced neuromuscular transmission and increased mitochondrial ATP production induced by photobiomodulation, although these mechanisms were not directly assessed.

Electrophysiological outcomes further supported the superiority of HILT. Distal motor latency improved from 5.3 ± 0.5 ms to 4.1 ± 0.4 ms in the HILT group, with smaller improvements in the LLLT group from 5.3 ± 0.4 ms to 4.5 ± 0.4 ms, and minimal change in control group from 5.4 ± 0.5 ms to 4.9 ± 0.5 ms. HILT patients demonstrated an increase in sensory nerve conduction velocity from 34.2 ± 3.6 m/s to 46.5 ± 3.8 m/s. The LLLT group demonstrated an increase from 33.8 ± 3.9 m/s to 43.9 ± 3.7 m/s, whereas the control group demonstrated an increase from 34.1 ± 3.5 m/s to 38.3 ± 3.4 m/s. The magnitude of improvement was greatest in the HILT group, followed by the LLLT group and the control group. These findings are consistent with previous studies reporting improved nerve conduction following HILT in patients with CTS [8, 16]. Variability in reports on LLLT effects may correlate with dosing and/or wavelength selection [26]. These electrophysiological improvements may be related to the observed increases in grip strength; however, the improvements seen in the control group also suggest that standardized physiotherapy and strengthening exercises may have contributed to functional gains.

The ultrasonographic analysis demonstrated a significant reduction in the CSA of the median nerve. The greatest reduction was found in the HILT group from 14.6 ± 1.8 mm^2 to 10.9 ± 1.4 mm^2. The LLLT group also showed a reduction in CSA from 14.4 ± 1.7 mm^2 to 11.7 ± 1.5 mm^2, whereas the control group showed a reduction from 14.5 ± 1.9 mm^2 to 13.1 ± 1.7 mm^2. These findings are consistent with those of Lin et al. who reported that laser therapy is associated with a reduction in median nerve CSA in patients with CTS. They further reported that reductions in CSA correlate with improvements in clinical and electrodiagnostic outcomes [27].

The differences observed between groups should be interpreted in the context of variations in treatment parameters, including fluence, wavelength, and power output. The HILT protocol delivered a higher energy density compared to the LLLT protocol; therefore, the observed effects cannot be attributed solely to differences in laser intensity. The findings reflect the comparative effectiveness of the specific photobiomodulation protocols used in this study. This study should be interpreted as a comparison of two clinically applied laser protocols rather than a direct mechanistic evaluation of laser intensity. The proposed biological mechanisms are based on existing literature and should be interpreted as hypotheses, as they were not directly measured in this study. It is suggested that future studies provide a complete report of irradiance, spot size and delivered energy for every treatment point to allow for better reproducibility.

Limitations

While the results are positive, the limitations of the study should be considered. The duration of the study was limited to six weeks, with immediate post-intervention assessments; therefore, the findings cannot be generalized to longer-term outcomes, recurrence rates, and continuing neural recovery. In addition, individuals with selected systemic or neurological conditions (e.g. hypothyroidism, myopathy, and myelopathy) were disqualified from the study in order to maintain sample homogeneity; therefore, these results may not fully apply to CTS populations with different distributions of systemic comorbidities. Although baseline characteristics such as BMI and occupational exposure were recorded, the present study was not designed to evaluate their independent influence on treatment response. Therefore, further studies are warranted to determine whether these factors modify responsiveness to HILT or LLLT in patients with CTS. Lastly, due to the fact that fixed laser parameters were used for both HILT and LLLT, there was no opportunity to evaluate the dose-response relationship or to develop individualized treatment protocols for HILT.

In addition, although outcome assessors were blinded, therapist blinding was not possible due to the nature of the intervention, which may have introduced performance bias. Furthermore, certain physical laser parameters were not directly verified; power output, irradiance, and spot sizes relied on manufacturer calibrations rather than direct independent measurements with standard power meters. Because actual delivered power density and beam profiles were not independently measured during the trial, potential minor variations in actual delivered irradiance could impact exact optical dose reproducibility and cross-study comparison of photobiomodulation parameters. Additionally, the study design compared two laser protocols that differed in multiple parameters, including wavelength, power output, and fluence; therefore, it was not possible to isolate the independent effect of laser intensity. The effectiveness of participant blinding was not formally evaluated, and differences in sensory perception between HILT, LLLT, and sham interventions (e.g., thermal effects) may have influenced participant awareness of treatment allocation. Furthermore, the study included only participants aged 20–40 years; therefore, the findings may not be directly generalizable to older adults with CTS, who may exhibit different clinical characteristics, comorbidity profiles, and responses to treatment. The trial was retrospectively registered on 17 March 2025, following the commencement of participant recruitment on 29 August 2024 due to administrative delays. Retrospective registration represents a methodological limitation that should be considered when evaluating the transparency of trial reporting.

Conclusion

High-intensity laser therapy demonstrated greater improvements than low-level laser therapy within the tested parameters, including pain intensity, functional status, grip strength, electrophysiological measures, and ultrasonographic assessment of median nerve cross-sectional area, in patients with subacute CTS. However, these findings reflect the comparative effectiveness of the specific treatment protocols and may be influenced by differences in fluence, wavelength, and power output rather than intensity alone. Both interventions were effective as adjuncts to physiotherapy. Further studies with longer follow-up are required to determine long-term clinical outcomes.

Key clinical implications

- Both HILT and LLLT, when used as adjuncts to traditional physiotherapy, are effective for improving clinical and functional outcomes in subacute CTS.
- The tested HILT protocol demonstrated greater improvements compared with the tested LLLT protocol over the 6-week period.
- These findings support the use of HILT as an adjunctive physiotherapy modality for short-term management of subacute CTS; however, treatment decisions should consider individual patient factors and protocol differences rather than assuming inherent superiority.

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Author contributions AAR, AM, & MS: Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work.AA, & SBT: Drafting the work or revising it critically for important intellectual content.AAR, AA, SBT, AM, & MS: Final approval of the version to be published.AAR: Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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